

Abhishek Sharma

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EDUCATION

- IOE Pulchowk Campus, Lalitpur, Nepal** Nov. 2019 – Apr. 2024
Bachelor in Computer Engineering 76% average
- SOSHGS, Gandaki, Pokhara, Nepal** 2017 – 2019
+2 Mathematics 3.85 GPA (CGPA 3.72)

EXPERIENCE

Founding Software Engineer April 2025 – Present
ZeroTB San Francisco, California

- Built core backend infrastructure and APIs from the ground up, contributing to overall system architecture and scalability decisions in an early-stage startup environment in compliance automation field.
- Engineered scalable database structure, optimized database performance, and executed large-scale data migrations and third-party integrations for production systems.
- Developed AI-powered workflows using RAG, LLM orchestration frameworks like LangChain, and LlamaIndex to enhance automation and intelligent data processing.
- Collaborated across product and engineering teams to rapidly prototype and ship backend features in a fast-paced startup setting.

Associate Software Engineer (ML) April 2024 – April 2025
Docsumo Chabahil, Kathmandu

- Contributed to the core document extraction pipeline for key-value pair and OMR extraction systems, including research, implementation, optimization, and refactoring of production APIs.
- Performed critical performance analysis on the key-value extraction pipeline and reduced extraction latency by ~ 27% (7-8 seconds), significantly improving system throughput and user experience.
- Worked closely with the SRE team to improve reliability and scalability of distributed systems using GCP, Kubernetes, Docker, and Redis.
- Built and maintained scalable backend services and APIs using Flask and MongoDB, supporting high-volume document processing workflows.

PROJECTS

Quasar: Linearizable Replicated Key-Value Store | [Link](#) | *Raft, SMR, WAL, FastAPI, Python*

- Implemented a Raft replicated state machine with term-based leader election, log-matching replication, and majority-quorum commit before applying writes.
- Built crash-durable recovery with a write-ahead log, snapshot compaction, and InstallSnapshot catch-up for lagging replicas under crash and partition.
- Served linearizable reads via a majority quorum check so an isolated leader cannot return stale state, and built an interactive lab to step through election, partition, crash, and stale-leader recovery on a live three-node cluster.

Topic Modeling for Nepali Political News (NLP) | [Link](#) | *LDA, BERTopic, nltk, Pandas, spaCy, gensim*

- Utilized Probabilistic, Algebraic, and Transformer-Based Methods for Topic Modeling in Low Resource Language like Nepali.
- Scraped and Cleaned data of Nepali language News from different online news portals.
- Paper Accepted at ICICT 2025 (To be Published in IEEE Xplore Post-Conference)

Videolize: Text Script to Video Generator | [Link](#) | *PyTorch, HuggingFace, Stable Diffusion, Gemini API*

- Developed deep learning model to convert sequential script to engaging video leveraging Stable Diffusion, Microsoft's Text to Audio model and Gemini API
- Images generated of N input scenes from stable diffusion is interpolated using spherical interpolation to ensure smooth transition.
- Project won first prize in GenAI category in Hack-A-Week Competition of Locus, Pulchowk Campus (2023)

Information Extraction from Invoices using LLM | *LLMs, YOLO, GPT4 API, Table-Transformers*

- Harnessed the power of Language Models (LLMs) to tackle the challenge of identifying tables and Key-Value pairs from provided invoice samples.
- Integrated the Vision GPT4 API for comprehensive language understanding while locally fine-tuning table transformers and YOLO (You Only Look Once) for precise object detection.
- This project won the first prize in LLM category in AI competition organized by Kathmandu University (2023)

Design of Network Infrastructure of a University | [Link](#) | *Cisco Packet Tracer*

- Designed a complex Network Topology of a sample University by including many departments.
- Did subnetting, included VLANs, Web and DNS Servers, Dynamic Routing by OSPF, etc.

Checkers | [Link](#) | *C++, CMake, Minimax, Alpha-beta pruning, SFML*

- Implemented Checkers board game using C++ in OOP, also AI (vs computer) was done
- Utilized AI algorithms in Game like Alpha-beta pruning which could beat any good human player.

Procedural Terrain Generator | [Link](#) | *C++, Core Graphics Algorithms, CMake*

- Created Terrain (Hills) Generator by using Procedural-generation algorithm (Perlin Noise)
- Could Modify the randomness in the algorithm, clipping amount, wireframe + non-wireframe, etc.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, SQL, Dart, HTML/CSS

Frameworks: Flask, FastAPI, Django, PyTorch, Node.js, Express.js, Flutter

Knowledge: Linux/Unix, Distributed Systems (Raft, SMR, linearizability), Computer Networks, Web Dev, Computer Vision (YOLO, CNN, ResNets, U-Net, etc.), Mathematical Domain (Linear Algebra, Probability and Statistics, etc.), NLP (RNNs, LSTMs), Transformer, DSA

Developer Tools: Git, Docker, Selenium, Postman, VS Code, Visual Studio, Cisco Packet Tracer

Libraries: gensim, spaCy, NLTK, scikit-learn, NumPy, Matplotlib, seaborn, Pandas

Databases: SQL (PostgreSQL, MySQL, SQLite), NoSQL (MongoDB)

Cloud Services: GCP, AWS (S3, EC2, Lambda, etc.)

AWARDS

Winner — LLM category in AI Competition organized by Kathmandu University [Dec. 2023] | [Link](#)

Winner — Generative AI category in LOCUS 2024 HackAWeek (Hackathon) [Jan. 2024]

Full Scholarship UnderGrad — **Rank 34** in IOE Examination (over 14000+ students) | [Link](#)

Rank 1 — In Entrance Examination of Electronics Department in Kathmandu University | [Link](#)

Golden Jubilee Scholarship — Awarded By Embassy of India in Kathmandu, Nepal

Position Holder — National Level Math Competition (Budanilkantha Math League) | [Link](#)

Entrance Topper — +2 level examination out of over 3000+ students

College Topper — +2 (Grade 12) Board Examination [**3.85 GPA**]

School Topper — SEE Level Examination [**3.8 GPA**]

CERTIFICATIONS

Machine Learning Specialization — Coursera & DeepLearning.AI | [Link](#)

Deep Learning Specialization — Coursera & DeepLearning.AI | [Link](#)

Annual Nepal AI School — NAAMII, Nepal | [Link](#)

AWS Academy Graduate - AWS Academy Cloud Architecting — AWS | [Link](#)

AWS Academy Graduate - AWS Academy Data Engineering — AWS | [Link](#)

COMMUNITY ENGAGEMENT

Secretary

Jan 2022 - May 2023

IT Club, Pulchowk Campus

- A representative from the Department of Electronics and Computer Engineering
- Collaborated with team to organize various tech events like Git Workshop, Design Workshop in College
- Increased the reach of the club by organizing tech events with participating students of other college too

Instructor

2023

LOCUS, Pulchowk Campus

- Collaborated on curriculum design
- Did teach over 100+ students in Software Fellowship of LOCUS
- Covered mathematical foundations and implemented ML algorithms from scratch and also using libraries

Software Programming Instructor

2021-2022

IT Club, Pulchowk Campus and PDSC, Pulchowk Campus

- Did teach over 250+ students in Workshop on Django Web Framework in PDSC
- Did teach over 40+ students physically in Workshop on Git and Github in IT Club